

Active Inference

Chapter 4: The Generative Models of Active Inference

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From Bayesian Inference to Free Energy

§ Bayes' Theorem

$$\Pr(x) \Pr(y | x) = \Pr(x | y) \Pr(y) \quad (1)$$

$$\Pr(y) = \sum_x \Pr(y, x) = \sum_x \Pr(y | x) \Pr(x) \quad (2)$$

Goal: model evidence, $\Pr(y)$, can be computed directly from the prior and likelihood

Problem: summation(or integration) to calculate evidence $\Pr(y) \rightarrow$ intractable!

Solution: convert it into an optimization problem

there is a relationship between the product of two probabilities and the sum of two information.

$$\underbrace{\Pr(x \cap y) = \Pr(x) \Pr(y)}_{\text{independence}} \implies \log(\Pr(x, y)) = \log(\Pr(x)) + \log(\Pr(y)) \quad (3)$$

thus, in Bayes' theorem, obtained prior information from prior and likelihood information

$$\underbrace{\log \Pr(x)}_{\text{log prior}} + \underbrace{\log \Pr(y | x)}_{\substack{\text{observation } y \\ \text{(log likelihood)}}} \longrightarrow \underbrace{\log \Pr(x | y)}_{\text{log posterior}} + \underbrace{\log \Pr(y)}_{\substack{\text{model} \\ \text{evidence}}} \quad (4)$$

From Bayesian Inference to Free Energy

Jensen's Inequality

Goal: convert equation for calculating $\Pr(y)$ into an optimization problem

Definition (Jensen's Inequality)

for convex function ϕ and random variable X , the following holds:

$$\mathbb{E}[\phi(X)] \geq \phi(\mathbb{E}[X]) \quad (5)$$

if ϕ is strictly convex, equality holds if and only if X is constant.

the negative logarithm function is strictly convex, so $\mathbb{E}[-\log(X)] \geq -\log(\mathbb{E}[X])$

$$\mathbb{E}[-\log(X)] \geq -\log(\mathbb{E}[X]) \Rightarrow \mathbb{E}[\log(X)] \leq \log(\mathbb{E}[X]) \quad (6)$$

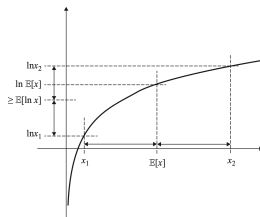


Figure 1: Logarithmic function providing intuition for Jensen's inequality.

From Bayesian Inference to Free Energy

Variational Free Energy

Goal: convert equation for calculating $\Pr(y)$ into an optimization problem

for approximate posterior $Q(x)$, we can obtain a lower bound of $\log \Pr(y)$:

$$\begin{aligned}\log \Pr(y) &\underbrace{=}_2 \log \sum_x \Pr(y, x) = \log \sum_x Q(x) \frac{\Pr(y, x)}{Q(x)} \\ &= \log E_{Q(x)} \left[\frac{\Pr(y, x)}{Q(x)} \right] \underbrace{\geq}_6 E_{Q(x)} \left[\log \frac{\Pr(y, x)}{Q(x)} \right] \triangleq -F[Q, y]\end{aligned}$$
$$E_{Q(x)}[\log \Pr(x, y)] = -F[Q, y] + E_{Q(x)}[\log Q(x)]: \text{ under approximate posterior}$$
$$F[Q, y] = D_{KL}[Q(x) \parallel \Pr(x | y)] - \log \Pr(y)$$
(7)

where $F[Q, y]$ is variational free energy.

we can rewrite Bayes' theorem to represent the **relationship between posterior and approximate posterior** $Q(x)$ (cf. 3rd line of Equation 7):

$$\begin{aligned}\Pr(x, y) &= \Pr(x | y) \Pr(y) \Rightarrow \\ \log \Pr(x, y) &= \log \Pr(x | y) + \log \Pr(y) \Rightarrow \\ E_{\Pr(x|y)}[\log \Pr(x, y)] &= \log \Pr(y) + E_{\Pr(x|y)}[\log \Pr(x | y)]: \text{ under true posterior}\end{aligned}$$
(8)

Equation 8 is satisfied because y is not a function of x (line 3) and expectation satisfies linearity.

Generative Models

to calculate the free energy, we need three things:

- data y
- a family of variational distributions $Q(x)$
- a generative model $\Pr(x, y)$

maybe, each kind of living things has a unique generative model: $P_{\text{cat}} \neq P_{\text{human}}$

the generative model used for Active Inference has two general forms:

- categorical sort
- continuous sort

Figure 2¹ shows the two basic forms of dynamic generative models used in Active Inference: a Partially Observable Markov Decision Process^{POMDP} and a continuous-time model.

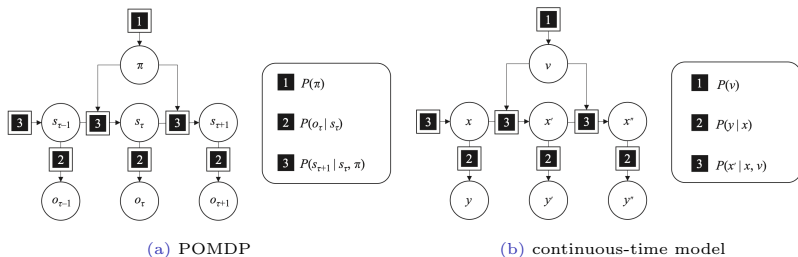


Figure 2: two dynamic generative models

¹the circles represent random variables, the squares represent the probability distributions

Generative Models

- Figure 2a, POMDP, which expresses a model in which a sequence of states (s) evolves over time. at each time step, states depend on the previous state and policy (π) selected by agent. each state generates an observation (o).
- Figure 2b, continuous-time model, which expresses a model in which a system's states² (x) evolve continuously over time according to some value v that plays a similar role to the policies. at each time point, states generate an observation (y).

we use different notation to emphasize the difference between categorical variables that evolve in **discrete time** (POMDP) and continuous variables that evolve in **continuous time** (continuous-time model).

- **discrete time:** s, π, o, P, Q ³
- **continuous time:** x, v, y, p, q ⁴

²to represent these values, we are going to use Taylor series expansion

³probability mass functions

⁴probability density functions

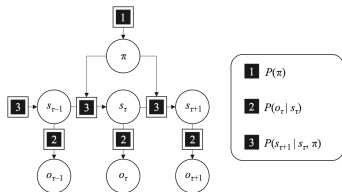
Active Inference in Discrete Time

in Figure 2a, a POMDP expresses the evolution over time of a sequence of hidden states that depend on a policy. we can convert it into a probabilistic model.

- **square 2:** $s_\tau \longrightarrow o_\tau$

$$P(o_\tau | s_\tau) = \text{Cat}(\mathbf{A})^a$$

$$\mathbf{A}_{ij} = P(o_\tau = i | s_\tau = j) \quad (9)$$



- **square 3:** $s_{\tau-1} \longrightarrow s_\tau \longrightarrow s_{\tau+1}$

$$P(\tilde{s} | \pi) = P(s_1) \prod_{\tau=1} P(s_{\tau+1} | s_\tau, \pi)$$

$$P(s_1) = \text{Cat}(\mathbf{D})$$

$$P(s_{\tau+1} | s_\tau, \pi) = \text{Cat}(\mathbf{B}_{\pi\tau}) \quad (10)$$

the term $P(\pi)$ in **square 1** can rewrite as:^aCategorical Distribution

$$P(\pi) = \text{Cat}(\boldsymbol{\pi}_0)$$

$$\boldsymbol{\pi}_0 = \sigma(-\mathbf{G})^5$$

$$\mathbf{G}_\pi = G(\pi) = -\mathbb{E}_{\tilde{Q}}[D_{KL}[Q(\tilde{s} | \tilde{o}, \pi) || Q(\tilde{s} | \pi)]] - \mathbb{E}_{\tilde{Q}}[\log P(\tilde{o} | \pi)] \quad (11)$$

$$\tilde{Q}(o_\tau, s_\tau | \pi) \triangleq P(o_\tau | s_\tau)Q(s_\tau | \pi)$$

where $\mathbf{G}_\pi$ is expected free energy under policy π .

⁵ σ : softmax function

Active Inference in Discrete Time

- **variational free energy**: prior data (y or o) $\longrightarrow$ adjust approximate posterior ($Q(x)$)
- **expected free energy**: no data (o) bcz no actions have been taken yet $\longrightarrow$ thinking... 'what data would I like to see?'

$$\begin{aligned} E_{\tilde{Q}}[D_{KL}[Q(\tilde{s} | \tilde{o}, \pi) || Q(\tilde{s} | \pi)]] \\ = -E_{Q(\tilde{o}|\pi)}[D_{KL}[Q(\tilde{s} | \tilde{o}, \pi) || Q(\tilde{s} | \pi)]] \end{aligned} \quad (12)$$

Note: we select policies that maximize the expected information gain (divergence) between beliefs updated with observations and those without.

$$\begin{aligned} G(\pi) &= -\underbrace{E_{\tilde{Q}}[D_{KL}[Q(\tilde{s} | \tilde{o}, \pi) || Q(\tilde{s} | \pi)]]}_{\text{information gain}} - \underbrace{E_{\tilde{Q}}[\log P(\tilde{o} | C)]}_{\text{pragmatic value}} \\ &= \underbrace{E_{Q(\tilde{s}|\pi)}[H[P(\tilde{o} | \tilde{s})]]}_{\text{expected ambiguity}} + \underbrace{D_{KL}[Q(\tilde{o} | \pi) || P(\tilde{o} | C)]}_{\text{risk}} \end{aligned} \quad (13)$$

where C is prior preferences. e.g., fish: 'i must be in water'.

Active Inference in Discrete Time

Goal: rewrite the expected free energy for efficient computation.

Abstract Definition

$$P(\pi) = \text{Cat}(\pi_0)$$

$$\pi_0 = \sigma(-\mathbf{G})$$

(concept)



Linear
Algebra

Linear Algebraic Form

$$\mathbf{G}_\pi = \mathbf{H} \cdot \mathbf{s}_{\pi\tau} + \mathbf{o}_{\pi\tau} \cdot \boldsymbol{\varsigma}_{\pi\tau}$$

$$\boldsymbol{\varsigma}_{\pi\tau} = \log \mathbf{o}_{\pi\tau} - \log \mathbf{C}_\tau$$

$$\mathbf{H} = -\text{diag}(\mathbf{A} \cdot \log \mathbf{A})$$

(calculation)

Note: this form allows implementing Active Inference using simple matrix-vector multiplications.

thus, the variational free energy F rewrites as:

$$F = \boldsymbol{\pi} \cdot \mathbf{F}$$

$$\mathbf{F}_\pi = \sum_{\tau} \mathbf{F}_{\pi\tau} \tag{14}$$

$$\mathbf{F}_{\pi\tau} = \mathbf{s}_{\pi\tau} \cdot (\log \mathbf{s}_{\pi\tau} - \log \mathbf{B}_{\pi\tau-1} \mathbf{s}_{\pi\tau-1}) - \mathbf{o}_{\pi\tau} \cdot \log \mathbf{A} \mathbf{s}_{\pi\tau}$$

Active Inference in Continuous Time

limit of a POMDP: sensory input and motor outputs are continuously evolving variables.

solution: continuous-time model (Figure 2b)